

Anzani Three-Cylinder Fan

Motorcycle engine manufacturer Alessandro Anzani's first aero engine might have been rough and ready—and even had one or two dubious design features—but it was light, produced just enough power, and was available at just the right time. Although Anzani went on to produce the first practical radial engine, his aeronautical star soon waned.

AVOIDING OVERHEATING

In design terms, Anzani already had the right architecture in place, in the form of the motorcycle engines he had built over many years. However, aircraft engines have to run for long periods at high power, and the consequent heat buildup causes problems. Anzani took the unusual approach of drilling holes around the base of the cylinders of his engine, which, when uncovered, allowed the early release of exhaust gases and an additional flow of cooling air at the bottom of the induction stroke. This curious remedy worked well enough to get Louis Blériot across the English Channel in 1909.

Paired cylinder holding-down nuts
Locking two nuts together kept them from vibrating loose.

Single-piece cast-iron cylinder
Cylinders were arranged at 60 degrees to each other in a fan configuration.

A single with two extra pots
While the Anzani fan engine looks like a section of a modern radial, on the inside it was very much a single-cylinder engine with an extra link rod and cylinder grafted on each side.

Two-blade, fixed pitch propeller
Carved from wood, this rotated at crankshaft speed (1,400-1,600 rpm).

Crankcase
Cast in aluminum to keep overall weight to a minimum.

